6.2.4 Biotic factors

The interactions among living organisms such as plants and animals are called **biotic factors,** which may cause marked effects upon vegetation. The effects may be direct and indirect and modifies the environment. The plants mostly which lives together in a community and influence one another. Similarly, animals in association with plants also affect the plant life in one or several ways. The different interactions among them can be classified into following two types they are positive interaction and negative interaction

Positive interactions

When one or both the participating species are benefited, it is positive interaction. Examples; Mutualism and Commensalism.

a. Mutualism: It is an interaction between two species of organisms in which both are benefitted from the obligate association. The following are common examples of mutualism.

Nitrogen fixation

Rhizobium (Bacterium) forms nodules in the roots of leguminous plants and lives symbiotically. The *Rhizobium* obtains food from leguminous plant and in turn fixes atmospheric nitrogen into nitrate, making it available to host plants.

Other examples:

- Water fern (*Azolla*) and Nitrogen fixing Cyanobacterium *(Anabaena*).
- *Anabaena* present in coralloid roots of *Cycas*. (Gymnosperm)
- Cyanobacterium (*Nostoc*) found in the thalloid body of *Anthoceros.*(Bryophytes)
- Wasps present in fruits of fig.
- Lichen is a mutual association of an **alga** and **a fungus**.
- Roots of terrestrial plants and fungal hyphae- **Mycorrhiza**

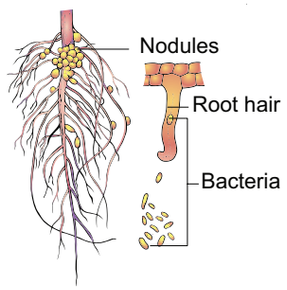


Figure 6.10: A nodulated legume plant root with bacteria

b. Commensalism: It is an interaction between two organisms in which one is benefitted and the other is neither benefitted nor harmed. The species that derives benefit is called the **commensal,** while the other species is called the **host**. The common examples of commensalism are listed below:

	Interaction type	Combination		Effects	Examples
1.Positive interaction					
1	**Mutualism**	(+)	(+)	Both species benefitted	Lichen, Mycorrhiza etc.
2	**Commensalism**	(+)	(0)	One species is benefitted and the other species is neither benefitted nor harmed	Orchids, Lianas etc.
2.Negative interaction					
4	**Predation**	(+)	(-)	One species benefitted, the other species are harmed	*Drosera, Nepenthes* etc.
5	**Parasitism**	(+)	(-)	One species benefitted, the other species are harmed	*Cuscuta, Duranta, Viscum* etc.
6	**Competition**	(-)	(-)	Harmful for both	Grassland species
7	**Amensalism**	(-)	(0)	Harmful for one, but the other species are unaffected	*Penicillium* and *Staphylococcus*

(+) Benefitted, (-) Harmed (0)Unaffected

Table 6.4: Different interactions of plant

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